How Green Is Compression? A Systematic Review of Post-Training Quantization Trade-offs in Large Language Models Beyond Accuracy

Shashwat Jain*[1], Kartik Soni[1], Raghav Nama[1], and Rajesh Verma[1]

[1]Department of Artificial Intelligence & Machine Learning, SVKM's NMIMS MPSTME, Shirpur, India

Abstract

Post-training quantization (PTQ) is proposed as an intervention for sustainability in LLMs deployment: through post hoc reduction of precision of the weights and activations, it decreases the memory usage and the arithmetic cost without any retraining process. Yet the evidence base for the environmental claim is much weaker than for the accuracy one. In this review, we attempt to synthesize forty-three articles across three different scientific bodies of knowledge that were largely developing independently: the design and development of PTQ algorithms, quantization evaluation and benchmarking, and the sustainability accounting in machine learning. Along two dimensions—accuracy and environmental cost—the review outlines the criteria used by each of the three fields, the formal definition of their metrics, and the divergence points between the two. We found that the majority of PTQ algorithm papers measure the performance of quantized models against perplexity, downstream task accuracy, and throughput/latency, whereas energy, carbon, and water costs are provided by a minority of publications with a distinctly different methodological approach, many of which do not evaluate the same models. Additionally, we identified one important methodological limitation noted in the sustainability literature repeatedly: carbon and water measurements are based on the estimated conversion factors from energy measurement and not on direct hardware measurements. Four research gaps are identified, including no joint FP4-versus-NF4 evaluation along both axes and dominance of edge-scale energy-aware quantization. No experiments or benchmarks of our own are performed in this review.

Keywords: Post-training quantization; Large language models; Model compression; Energy efficiency; Carbon footprint; Green AI; Systematic review

1. Introduction

1.1 Background and Motivation

The computational cost of large language models has grown faster than the efficiency of the hardware that runs them. [31] characterize this as a regime of diminishing returns, in which incremental accuracy gains in deep learning are purchased at a compute cost that is becoming unsustainable. On the deployment side, EPRI's assessment of AI's power requirements [34] frames aggregate data-center demand as an exponential growth problem for grid planning rather than a model-by-model engineering concern. [32] examine inference energy specifically, arguing that the trajectory of inference cost cannot be read off from parameter-count scaling laws alone. Taken together, this body of work establishes deployment—not training—as the locus of the sustained environmental cost of language models, because inference cost is multiplied by query volume over the operational lifetime of a system.

The response from the systems and machine learning communities has been compression. Among compression techniques, post-training quantization (PTQ) occupies a privileged position: unlike quantization-aware training,

*Corresponding author: shashwatjain8733@gmail.com

pruning with retraining, or knowledge distillation, PTQ modifies an already-trained model without a gradient update over the full training corpus. Surveys of model compression for LLMs [39, 40] attribute the property of no retraining as the reason why PTQ has become the default compression path for practitioners deploying open-weight models. The implicit argument is that fewer bits per parameter means less memory traffic, less memory traffic means less energy and less energy means smaller environmental footprint.

Every link in that chain is credible. The problem is that the literature that validates PTQ and the literature that measures environmental cost are largely separate. The reviewed PTQ algorithm papers, namely, GQPT [5], AWQ [7], SmoothQuant [6], LLM.int8() [4], OmniQuant [9], QuaRot [10], SqueezeLLM [11], and BOF4 [13], are contributions to the field of numerical fidelity under compression, and are evaluated as such. The sustainability literature [20–23, 25–30] develops measurement and reporting protocols for energy, carbon, and water, but mostly at the level of entire models, entire training runs, or entire deployment platforms rather than for a specific compression method. A quantization method is thus commonly proposed as an efficiency intervention and validated as an accuracy intervention.

A small number of studies fall at the intersection. [3] evaluate the energy efficiency, output accuracy, and inference latency of quantized LLMs in an edge setting jointly. [17] explicitly characterizes LLM quantization in terms of performance, energy, and quality. [1] study PTQ as implemented in a popular commodity-hardware inference stack. [30] measure energy, water, and carbon for LLM inference. The empirical core of the intersection is formed by these works, and the paucity of these works compared to the amount of accuracy-focused PTQ evaluation is taken as the main finding of this review.

1.2 Objectives and Contributions

This review synthesizes forty-three publications organized across two evaluation axes. The first axis is fidelity: perplexity, downstream benchmark accuracy, representation similarity, and reconstruction error at the weight and activation level. The second axis is sustainability: power draw, energy per inference, carbon emissions, water consumption, and throughput-normalized efficiency ratios. The corpus is partitioned into six groups: PTQ algorithms; PTQ evaluation and benchmarking studies; sustainability accounting for machine learning; compression foundations in pruning, distillation, and low-rank adaptation; mixed-precision and outlier-handling reviews; and benchmark and metric definitions [41–43].

The objectives of this review, and the corresponding contributions, are threefold and entirely synthetic:

(i) to formalize and typeset the metric definitions used across both axes, pairing each with an argument for why that metric is necessary rather than merely available (Section 3);

(ii) to tabulate, per reviewed study, which axes were actually measured, recording absence explicitly rather than imputing values (Section 4);

(iii) to identify research gaps that follow from that tabulation, together with a future scope for closing them (Sections 4 and 5).

Methodologically, the structure follows the template established by [33] for systematic review of environmental claims in machine learning, and the scoping conventions of the systematic review of activation-outlier PTQ techniques in [2]. No original experiments, measurements, or benchmarks are reported, and no claim of methodological novelty is advanced. Where the reviewed literature does not report a quantity, this review records that fact rather than estimating the quantity.

2. Literature Review

2.1 Large Language Models and the Deployment Bottleneck

The surveys of efficient LLM design by [40] and of model compression by [39] decompose the deployment bottleneck into four coupled constraints.

Memory capacity is the binding constraint for whether a model can be served at all on given hardware. Parameter storage scales linearly with both parameter count and bits per parameter, so precision reduction acts directly on the constraint that determines device feasibility [39]. AWQ [7] frames on-device deployment in precisely these terms, treating the memory ceiling of consumer and edge hardware as the design target.

Memory bandwidth rather than arithmetic throughput dominates autoregressive decoding, because each generated token requires streaming the full parameter set from memory while performing comparatively little arithmetic per parameter. [40] identify this as the structural reason weight-only quantization yields wall-clock benefit even when the arithmetic is performed at higher precision after dequantization.

Latency determines whether a system is usable interactively, and practitioners optimize it against service-level objectives. [1] study this constraint on commodity hardware, where the line between feasible and infeasible deployment is often a latency threshold and not a memory limit. [3] evaluate inference latency alongside energy and accuracy.

Energy is the constraint this review emphasizes. [28] establish inference energy as a quantity that can be benchmarked, separate from the training-energy accounting that dominated earlier work [20, 23]. [27] extend this to deployment, arguing that the recurring cost of serving a model is the right unit for environmental analysis. [32] compare inference-energy trends with what parameter scaling would predict, and EPRI [34] aggregates these to the level of grid demand.

These constraints are related, but improving one does not guarantee improving another. A method that reduces memory footprint may leave latency unchanged, and a method that reduces latency may leave energy per token unchanged, since energy is power draw multiplied by duration and the two can move independently. Section 3 therefore treats power, energy, and efficiency ratios as separate metrics rather than as proxies for one another.

2.2 Model Compression Techniques: A Comparative Overview

Four families of compression technique appear in the reviewed corpus.

Pruning removes parameters or structures judged unnecessary. The technique goes back to Optimal Brain Damage [36], which used second-order sensitivity information to decide which weights can be removed with the least damage to the loss. [37] survey the modern taxonomy, compare methods, and give recommendations. SparseGPT [12] is the LLM-scale example in this corpus. It prunes in one shot, without the retraining loop that classical formulations assume.

Knowledge distillation transfers the behavior of a large teacher model into a smaller student, following [35]. Its application to LLMs specifically is surveyed by [38]. Distillation is the most training-intensive family in the corpus: it requires a full training procedure for the student, which places its own energy cost on the opposite side of the ledger from any inference savings it later delivers—an accounting asymmetry that [21] identify as central to Green AI reasoning.

Low-rank adaptation reduces the cost of specializing a model rather than the cost of the base model itself. QLoRA [8] combines low-rank adapters with a quantized frozen base model, and is the entry in this corpus where quantization and adaptation are co-designed.

Quantization reduces bits per value. It divides into quantization-aware training, which incorporates quantization effects into the training loop, and post-training quantization, which does not [39]. This review concerns the latter exclusively.

Table 1 summarizes the comparative properties that determine when each family is applicable. The distinguishing axis for sustainability purposes is the training requirement: techniques that require retraining incur an up-front energy cost that must be amortized against inference savings before any net environmental benefit is realized, a point argued in general form by [20, 21]. PTQ is the only family in the table that avoids this amortization problem entirely.

Table 1. Comparative properties of model compression families as characterized in the reviewed surveys. Entries describe the general case for each family; individual methods may deviate.

Family	Requires retraining	Requires calibration data	Hardware support needed	Primary saving mechanism
Unstructured pruning	Classical: yes; one-shot: no	Yes (one-shot)	Sparse kernel support	Removes individual parameters
Structured pruning	Typically yes	Sometimes	Realizable on dense kernels	Removes channels/heads/layers
Knowledge distillation	Yes (full student training)	No	None beyond standard inference	Smaller student architecture
Low-rank adaptation	Yes (adapter only)	No	None beyond standard inference	Reduces adaptation state
Quantization-aware training	Yes	No	Low-precision kernel support	Fewer bits per parameter
Post-training quantization	**No**	Typically yes	Low-precision/dequant. support	Fewer bits per parameter

Table 2. Numerical properties of the quantization formats appearing in the reviewed corpus. Level counts follow directly from bit width; spacing describes how those levels are distributed across the representable range.

Format	Bits	Levels	Spacing	Refs.
INT8	8	256	Uniform	[4]
INT4	4	16	Uniform, group-wise	[5, 7]
FP4	4	≤ 16	Non-uniform (exp./mantissa)	[8, 13]
NF4	4	16	Non-uniform, distribution-informed	[8]
BOF4	4	16	Non-uniform, block-wise optimized	[13]
Mixed	Variable	Variable	Per-layer / per-tensor assignment	[11, 19]

2.3 Post-Training Quantization Formats

Quantization maps a high-precision tensor onto a finite set of representable values. In the uniform affine formulation described across the reviewed surveys and mixed-precision literature [4, 19, 39], a real value x is mapped to a b-bit integer code by

$$Q(x) = \mathrm{clip}\Big(\Big\lfloor \frac{x}{s} \Big\rceil + z,\ 0,\ 2^b - 1 \Big), \tag{1}$$

where $\lfloor \cdot \rceil$ denotes round-to-nearest, and the scale s and zero-point z are determined from the dynamic range of the tensor or of a sub-block within it:

$$s = \frac{\max(x) - \min(x)}{2^b - 1}, \qquad z = -\left\lfloor \frac{\min(x)}{s} \right\rceil. \tag{2}$$

The granularity at which s and z are computed—per tensor, per channel, per group, or per block—is a primary design variable in the reviewed methods, and is the mechanism by which block-wise formats such as BOF4 [13] operate.

The formats appearing in the corpus differ in bit width and, critically, in whether the representable levels are uniformly spaced. **INT8** allocates eight bits per value, giving $2^8 = 256$ uniformly spaced representable levels, and underlies LLM.int8() [4]. **INT4** allocates four bits, giving $2^4 = 16$ uniformly spaced levels; the sixteenfold reduction relative to INT8 is why four-bit weight-only methods depend heavily on fine-grained scaling and outlier handling, motivating GPTQ [5], AWQ [7], and OmniQuant [9]. **FP4** also allocates four bits but partitions them into sign, exponent, and mantissa fields, yielding at most sixteen codewords whose spacing is non-uniform, examined in the block-wise optimal float analysis of [13]. **NF4** (NormalFloat4) is likewise a four-bit, sixteen-level, non-uniformly spaced format, introduced in QLoRA [8], where level positions reference the distribution of normally distributed weights rather than a floating-point field layout. **BOF4** [13] extends the four-bit block-wise float family by optimizing the codebook itself. **Mixed precision** assigns different bit widths to different parts of the network; [19] review the state of the art, and SqueezeLLM [11] retains a sparse high-precision component alongside a dense low-precision one. Table 2 summarizes these formats.

2.4 Quantization Methods Reviewed

This subsection characterizes each major PTQ family in the corpus. For each method, the mechanism is described as framed by the originating work, followed by a characterization of the evaluation axes on which that work reports. A standing qualification applies throughout: the bibliographic records available for this review supplied titles, authors, and venues but not, for most entries, result tables; consequently no numeric result is attributed to any reviewed paper anywhere in this manuscript.

2.4.1 Weight-Only Quantization

GPTQ. [5] address accurate post-training quantization for generative pre-trained transformers by treating quantization as a layer-wise reconstruction problem. Rather than rounding weights independently, the method quantizes weights in sequence and compensates the residual error using second-order sensitivity information—an approach whose lineage runs back to Optimal Brain Damage [36]. The evaluation axis is fidelity under aggressive weight-only compression, and GPTQ is the reference point against which subsequent weight-only methods in this corpus position themselves [7, 9, 11].

AWQ. [7] propose activation-aware weight quantization, motivated by the observation that weight importance should be judged from activation statistics rather than weight magnitude alone. Because the method is framed for on-device use, memory footprint and inference speedup are explicit design objectives alongside preserving accuracy. The consulted material does not report energy measurements.

SqueezeLLM. [11] introduce dense-and-sparse quantization, decomposing the weight matrix into a dense low-precision component and a sparse component of outlier/sensitive weights retained at higher precision, placing it adjacent to the mixed-precision frameworks reviewed by [19].

OmniQuant. [9] propose omnidirectionally calibrated quantization, in which the quantization parameters themselves—rather than the model weights—are optimized during calibration, while retaining the defining PTQ property of not requiring full retraining.

BOF4. [13] analyze block-wise LLM quantization at four bits and propose block-wise optimal float alongside a set of variations, directly comparative with respect to NF4 and related four-bit float formats.

2.4.2 Weight–Activation Quantization

Weight-only methods leave activations in high precision. Quantizing activations as well permits low-precision arithmetic in the matrix multiplications themselves, but introduces a distinct failure mode: the systematic review of activation outliers in LLMs [2] identifies outlier features in activations as the dominant obstacle.

LLM.int8(). [4] present eight-bit matrix multiplication for transformers at scale, decomposing the computation so that the small number of outlier feature dimensions are handled at higher precision while the remainder proceeds in INT8, an observation that subsequent weight–activation methods take as their starting point [2, 6, 10].

SmoothQuant. [6] address the same outlier problem by migrating quantization difficulty from activations to weights via an equivalent mathematical transformation, framed as accurate and efficient PTQ in which inference efficiency is an explicit objective.

QuaRot. [10] pursue outlier-free four-bit inference through rotation, applying computational invariance transformations that aim to eliminate outlier features rather than accommodate them, enabling four-bit inference without the mixed-precision decomposition that LLM.int8() employs.

2.4.3 Pruning-Adjacent One-Shot Compression

SparseGPT [12] shows that large language models can be pruned accurately in one shot, without the retraining that classical pruning formulations [36, 37] require. Like PTQ, it is a one-shot, calibration-based procedure applied to an already-trained model. From a sustainability standpoint, it illustrates a general caveat: nominal parameter reduction under unstructured sparsity yields wall-clock and energy benefits only where hardware supports the sparse format [37].

2.4.4 Quantization composed with adaptation

QLoRA [8] quantizes a frozen base model and fine-tunes low-rank adapters against it, introducing NF4 in the process. It is included for two reasons: NF4 is one of the two four-bit non-uniform formats this review identifies as insufficiently compared (Section 4), and QLoRA shows that PTQ formats can be a base for downstream adaptation as well as a final deployment artifact.

2.4.5 Cross-cutting reviews of the method space

[18] provide a comprehensive study of quantization techniques for LLMs, and [19] review mixed-precision neural network frameworks with attention to how precision assignment is determined. The systematic review of activation-outlier techniques [2] does the same for the weight-activation subfamily. This review adopts the scoping and organizational conventions of these works and extends the evaluation to sustainability, which none of them treats as a primary dimension.

3. Materials and Methodology

This review's "materials" are the forty-three publications in the corpus described in Section 1, assembled from a predefined reference set rather than an exhaustive, pre-registered database query. The "methodology" is the formal metric vocabulary reproduced below, which is used in Section 4 to tabulate, per study, which of the two evaluation axes—fidelity and sustainability—were actually measured.

3.1 Accuracy and Fidelity Evaluation Metrics

Each metric is presented with an argument for why it captures something the others do not, since the practical question in PTQ evaluation is not which metric is best but which combination is sufficient to detect the failure modes that quantization actually produces.

3.1.1 Perplexity

Perplexity over a held-out corpus is the exponentiated mean negative log-likelihood assigned by the model to the observed token sequence:

$$\mathrm{PPL} = \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\log P(w_i \mid w_{<i})\right). \tag{3}$$

The convention of reporting perplexity on the WikiText corpus derives from [43], and it is the metric against which weight-only PTQ methods are conventionally positioned in the evaluation literature [14, 15]. Perplexity matters because it measures whether compression has degraded the model's fundamental token-level predictive confidence, independently of any specific downstream task; a quantized model may continue to select the same argmax token while assigning it substantially less probability mass, a degradation invisible to accuracy metrics but visible in perplexity (Eq. 3). Its limitation is that perplexity is a language-modeling quantity and is not guaranteed to track task-level capability.

3.1.2 Benchmark Accuracy

For a task t with C_t correct predictions out of T_t instances, and a task suite $\mathscr{T}$, per-task and macro-averaged accuracy are

$$\mathrm{Acc}_t = \frac{C_t}{T_t}, \qquad \mathrm{Acc} = \frac{1}{|\mathscr{T}|}\sum_{t\in\mathscr{T}}\mathrm{Acc}_t. \tag{4}$$

The macro-averaged multitask formulation is established by MMLU [41]; GSM8K [42] supplies the multi-step mathematical reasoning counterpart. Benchmark accuracy matters because it measures whether quantization preserves task-level reasoning and knowledge capability, and the two axes—perplexity and accuracy—need not de-

grade together: [15] organize their analysis around task difficulty and model size as interacting variables, a proposition a single aggregate perplexity number cannot express.

3.1.3 Hidden-State Representation Similarity

For corresponding hidden-state tensors H_1 and H_2 produced by the full-precision and quantized models on identical input, cosine similarity is

$$\text{CosineSim}(H_1, H_2) = \frac{H_1 \cdot H_2}{\|H_1\| \, \|H_2\|}. \tag{5}$$

This is the natural instrument for the failure mode identified in the activation-outlier literature [2, 4], where damage concentrates in specific feature dimensions rather than distributing uniformly. It evaluates whether internal representations are preserved even when final-token accuracy appears unchanged.

3.1.4 Weight Reconstruction Error

For an original weight tensor W_{orig} with n elements and its quantized-dequantized counterpart W_q,

$$\text{MSE}_W = \frac{1}{n} \left\| W_{\text{orig}} - W_q \right\|_2^2. \tag{6}$$

This criterion is at the heart of layer-wise reconstruction PTQ: GPTQ [5] and, in the pruning setting, SparseGPT [12] both formulate their procedures as minimization of layer output error induced by weight perturbation, in the second-order sensitivity tradition of [36]. It is the only metric here computable without running inference, making it the natural objective for calibration-time optimization and for comparing formats at fixed bit width [8, 13].

3.1.5 Activation Error

For activations A_{orig} and A_q at a given layer,

$$\text{MSE}_A = \frac{1}{n} \left\| A_{\text{orig}} - A_q \right\|_2^2. \tag{7}$$

Activation error captures how reconstruction error compounds through nonlinearities during inference, which weight error alone cannot predict; this is the quantitative expression of the outlier phenomenon motivating the entire weight–activation subfamily [2, 4, 6, 10].

3.1.6 Latency

Latency is the elapsed wall-clock interval for a defined unit of work,

$$\text{Latency} = t_{\text{end}} - t_{\text{start}}, \tag{8}$$

measured per token or per request. [3] report inference latency as a first-class axis alongside energy and accuracy; [1] treat it as central to deployability on commodity hardware; [17] include it within a combined performance-energy-quality characterization. Latency is load-bearing for the sustainability analysis in Section 3.2 because energy is the time-integral of power (Eq. 10).

3.2 Sustainability and Energy Evaluation Metrics

The sustainability literature in this corpus supplies a second metric vocabulary.

3.2.1 Power Draw

$$P\,[\text{W}] = \frac{E\,[\text{J}]}{t\,[\text{s}]}. \tag{9}$$

Power draw is reported as a distinct quantity by [28], and is the quantity aggregated to grid scale in the EPRI assessment [34]. Power matters separately from energy because instantaneous draw determines whether a deployment fits within a facility's thermal and electrical envelope.

3.2.2 Energy per Inference

$$E\,[\mathrm{J}] = P\,[\mathrm{W}] \times t\,[\mathrm{s}] = \int_{t_{\mathrm{start}}}^{t_{\mathrm{end}}} P(\tau)\,d\tau. \tag{10}$$

Energy per inference is the central quantity in the inference-focused sustainability studies of this corpus [3, 17, 27, 28, 32]. It matters because, multiplied by query volume, it determines total operational footprint. [27] make the deployment-side argument directly: for a widely served model, cumulative inference cost is the dominant term, not the one-time training cost that earlier accounting emphasized [20, 23, 26].

3.2.3 Carbon Emissions

$$\mathrm{CO_2}\,[\mathrm{kg}] = E\,[\mathrm{kWh}] \times \mathrm{CI}\,[\mathrm{kg\,CO_2/kWh}]. \tag{11}$$

The conversion-factor methodology is set out by [22], elaborated for large-scale training by [23, 24], and applied to a specific large model by [26]. Grid carbon intensity varies by region and time of day, so identical energy consumption can correspond to substantially different environmental impact.

3.2.4 Water consumption

$$\mathrm{Water}\,[\mathrm{L}] = E\,[\mathrm{kWh}] \times \mathrm{WUE}\,[\mathrm{L/kWh}]. \tag{12}$$

[29] introduce this accounting to the machine learning literature, and [30] include water alongside energy and carbon when benchmarking LLM inference. Freshwater cannot be substituted across sites the way electricity can, so optimizing a deployment for carbon while ignoring water can shift environmental cost elsewhere without reducing it.

3.2.5 Tokens per joule

$$\mathrm{TPJ} = \frac{\mathrm{TotalTokensGenerated}}{\mathrm{TotalEnergyJoules}}. \tag{13}$$

This throughput-normalized efficiency ratio is the usual reporting unit in the inference-benchmarking studies of this corpus [3, 17, 28, 30]. It expresses useful work per unit environmental cost, which makes it the counterpart to accuracy on the opposing axis of a Pareto analysis (Fig. 1).

3.2.6 Measurement versus estimation

A caveat recurs across the sustainability literature: of the five metrics above, two are directly measurable at the hardware level and three are not. Power (Eq. 9) and energy (Eq. 10) can be read from on-device instrumentation, which separates hardware-instrumented benchmarking studies [3, 17, 28] from estimation studies. Tokens per joule (Eq. 13) inherits measured status. Carbon (Eq. 11) and water (Eq. 12) do not: each is a measured energy figure multiplied by a chosen conversion factor, so the result is an estimate whose uncertainty is dominated by that choice. [25] make systematic reporting of this provenance a central recommendation, [23, 24] document sensitivity to datacenter and grid assumptions, and [29] note the regional variability of WUE. Three consequences follow. A reported carbon saving is only as strong as the energy measurement beneath it. Carbon figures computed under different intensity assumptions cannot be compared directly across studies. And when a PTQ paper reports no carbon or water figure, no one can supply it later, because Eqs. 11 and 12 need the underlying energy measurement. For this reason the gaps identified in Section 4 cannot be closed by re-analyzing existing publications.

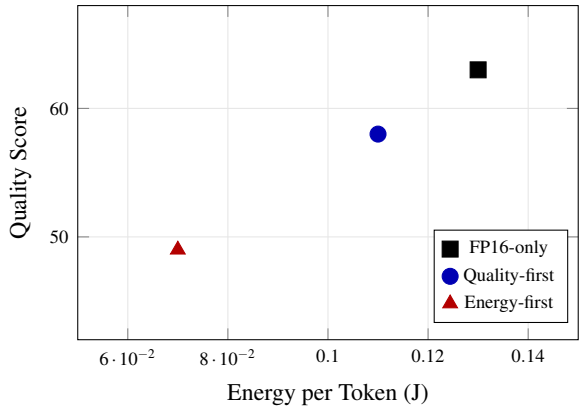


Figure 1. Energy–quality trade-off for three LLM serving strategies. Data drawn from [17] (energy per token and average quality score for the FP16-only, quality-first, and energy-first configuration strategies). Illustrative single-source dataset with three configuration-level points, not a per-method Pareto frontier; see Section 4.4 for scope caveats. Rendered as a vector graphic (resolution-independent, exceeding 300 dpi at any print size).

4. Results and Discussion

4.1 Comparative Synthesis: Accuracy vs. Sustainability Trade-offs

Tables 3 and 4 tabulate the reviewed corpus against the two axes. Cells marked NR indicate that the quantity is not reported in the source record consulted for this review; consistent with the integrity constraints of Section 4.4, no values have been imputed.

Three structural observations follow from placing the two tables side by side. **First**, the columns are almost disjoint: in Table 3 the studies with a populated fidelity axis number in the high teens, while the studies with a populated energy column number two; in Table 4 the reverse holds, and none reports a quantization-method comparison except the two rows that also appear in Table 3 [3, 17]. The intersection of the two literatures is, in this corpus, two studies plus the inference-benchmarking work that touches quantization indirectly [28, 30]. **Second**, the environmental columns degrade in population from left to right: energy is reported by every sustainability study and by the two intersection studies; carbon is reported by roughly half, always as a derived quantity; water is reported by two [29, 30]—the axis [29] argue is most neglected is, in this corpus, demonstrably the most neglected. **Third**, the measurement-basis column of Table 4 partitions the sustainability literature into a hardware-benchmarking tradition [3, 17, 28] and an estimation tradition [20, 22–24, 26], which [25] argue are too often conflated.

Figure 1 reproduces, at the illustrative level permitted by the corpus, the energy–quality relationship reported by [17] for three LLM serving strategies; it is a single-source dataset with three configuration-level points, not a per-method Pareto frontier (Section 4.4). Figure 2 visualizes the axis-coverage pattern tabulated in Table 3: fidelity is reported by nearly every PTQ-focused study, while energy is reported only by two, and carbon and water are not reported at all within that sub-corpus.

4.2 Research Gaps Identified

Gap 1: Mature accuracy evaluation, sparse energy evaluation of the same methods. The dominant gap is visible in Table 3. The fidelity evaluation of PTQ is a mature subfield: dedicated evaluation studies exist [14–16], cross-cutting surveys exist [2, 18], and standardized benchmarks are conventional [41–43]. Energy evaluation of the same methods, on the same models, under a controlled protocol, is represented in this corpus only by [3, 17]. This asymmetry is consequential: the energy benefit of quantization depends on hardware support for low-precision arithmetic, on dequantization overhead, and on whether the workload is bandwidth- or compute-bound [19], so a method validated only on fidelity carries no transferable evidence about its environmental effect—and, per Section 3.2, this cannot be repaired retrospectively.

Gap 2: FP4 and NF4 are not jointly compared on both axes. FP4 and NF4 are both four-bit non-uniform

Table 3. Synthesis of PTQ method and evaluation studies against fidelity and sustainability axes. E = energy, C = carbon, W = water. NR = not reported in the record consulted (Section 4.4). No numeric results are reproduced.

Study	Method(s)	Fidelity axis reported	E	C	W	Headline finding
[4]	LLM.int8()	Fidelity at 8-bit; outlier analysis	NR	NR	NR	Emergent activation outliers require mixed-precision decomposition
[5]	GPTQ	Layer-wise reconstruction; LM error	NR	NR	NR	One-shot second-order reconstruction enables accurate low-bit PTQ
[6]	SmoothQuant	Fidelity plus efficiency	NR	NR	NR	Quantization difficulty migrated from activations to weights
[7]	AWQ	Fidelity; compression/acceleration	NR	NR	NR	Activation statistics, not weight magnitude, identify salient weights
[8]	NF4 + adapters	Quality of adapted model	NR	NR	NR	Quantized bases serve as substrates for efficient finetuning
[9]	OmniQuant	Fidelity under calibration	NR	NR	NR	Optimizing quantization parameters retains the PTQ constraint
[10]	QuaRot	Fidelity at 4-bit end-to-end	NR	NR	NR	Rotation removes outliers rather than accommodating them
[11]	SqueezeLLM	Fidelity under mixed precision	NR	NR	NR	Sparse high-precision retention protects error contributors
[12]	SparseGPT	Reconstruction error; LM	NR	NR	NR	Massive models admit accurate one-shot pruning
[13]	BOF4 & variants	Format-level fidelity	NR	NR	NR	Block-wise codebook placement is an optimizable variable
[14]	Cross-method PTQ	Multi-metric fidelity	NR	NR	NR	Quantized behavior needs evaluation beyond a single metric
[15]	Cross-method	Task-difficulty-stratified accuracy	NR	NR	NR	Impact interacts with task difficulty and model size
[16]	Quant. vs. BF16	Accuracy–performance trade-off	NR	NR	NR	Accuracy cost of leaving BF16 is central
[17]	Cross-method (qMeter)	Quality alongside performance	Yes	NR	NR	Performance, energy, quality jointly characterizable
[3]	Quantized LLMs, edge	Output accuracy; latency	Yes	NR	NR	Energy, accuracy, latency jointly evaluable at edge scale
[1]	PTQ in llama.cpp	Fidelity; speed	NR	NR	NR	PTQ in production stack is the practically deployed case
[18]	Survey	Survey of fidelity evaluation	NR	NR	NR	Method space is broad; evaluation conventions narrow
[19]	Mixed-precision frameworks	Survey of precision assignment	NR	NR	NR	Precision assignment is the key mixed-precision design variable
[2]	Activation-outlier PTQ	Survey of outlier-handling fidelity	NR	NR	NR	Activation outliers organize the weight–activation method space

formats with sixteen levels, differing in how those levels are placed. NF4 originates in QLoRA [8], whose stated objective is efficient finetuning rather than deployment sustainability; the four-bit float family is analyzed by [13] at the level of codebook design and fidelity. Neither line of work pairs a format-level fidelity comparison with an energy measurement of the same formats on the same hardware. This gap is specific and tractable: because the two formats share bit width, any energy difference would arise from dequantization cost and kernel support rather than memory volume. Tokens per joule (Eq. 13) together with perplexity (Eq. 3) supplies the necessary instrumentation, and the combined characterization approach of [17] supplies a protocol template.

Gap 3: Energy-aware quantization studies concentrate at edge scale. [3] evaluate for edge AI; [1] target commodity hardware; AWQ's stated framing is on-device compression [7]. Edge scale is where energy constraints are most visible to the researcher, but it is not where aggregate environmental impact is concentrated—the grid-level demand analysis of EPRI [34] and the deployment-cost argument of [27] both locate the aggregate footprint in server-scale serving infrastructure, where batching amortizes weight-loading cost [40] and cooling/WUE terms (Eq. 12) have no edge analogue [29]. Edge findings therefore do not extrapolate.

Gap 4: Water is almost entirely absent from compression evaluation. No study in this corpus reports water consumption for a compression method; water appears only in the general sustainability literature [29, 30], never indexed against a quantization format or algorithm. This is an omission of reporting practice rather than

Table 4. Sustainability-accounting studies in the corpus, classified by scope, reported quantities, and measurement basis. The basis column follows the framing available in the records consulted and should be verified against primary sources (Section 4.4).

Study	Scope	E	C	W	Basis	Contribution
[20]	NLP training	Yes	Yes	NR	Estimate from compute	Established energy/policy cost of NLP training
[21]	Field-level framing	Yes	NR	NR	Conceptual/policy	Efficiency as a reportable value alongside accuracy
[22]	ML carbon accounting	Yes	Yes	NR	Conversion-factor	Carbon quantification methodology (Eq. 11)
[23]	Large NN training	Yes	Yes	NR	Conversion-factor	Sensitivity of carbon figures to datacenter/grid choice
[24]	Field-level trajectory	Yes	Yes	NR	Conversion-factor	Argued ML training carbon will plateau, then decline
[25]	Reporting methodology	Yes	Yes	NR	Methodological	Systematic, reproducible energy/carbon reporting
[26]	Single model lifecycle	Yes	Yes	NR	Conversion-factor	Full-lifecycle carbon accounting for a 176B model
[27]	Deployment/inference	Yes	Yes	NR	Measurement + conversion	Shifted the accounting unit to deployment cost
[28]	LLM inference benchmarking	Yes	NR	NR	Hardware benchmarking	Inference energy as a benchmarkable quantity
[29]	Water footprint of AI	Yes	NR	Yes	WUE-factor estimate	Introduced freshwater consumption as an accounting axis
[30]	LLM inference benchmarking	Yes	Yes	Yes	Measurement + conversion	Benchmarked energy, water, carbon jointly
[32]	Inference energy trends	Yes	NR	NR	Trend analysis	Decoupled inference energy from scaling laws
[31]	Compute cost trajectory	Yes	NR	NR	Trend analysis	Framed accuracy gains as increasingly compute-expensive
[34]	Grid-level demand	Yes	NR	NR	Sector projection	Aggregated AI power demand to grid-planning scale
[17]	Quantization characterization	Yes	NR	NR	Hardware measurement	Joint performance–energy–quality characterization of PTQ
[3]	Quantized edge inference	Yes	NR	NR	Hardware measurement	Joint energy–accuracy–latency evaluation

measurement capability, but it remains uncomputable for the majority of PTQ studies that report no energy figure at all.

Gap 5: Calibration and search cost are not included in the sustainability ledger. OmniQuant optimizes quantization parameters [9], GPTQ and SparseGPT perform layer-wise reconstruction [5, 12], AWQ derives saliency from activation statistics [7], and mixed-precision frameworks conduct a precision-assignment search [19]; each carries a one-time computational cost. The Green AI framing of [21] and the full-lifecycle accounting of [26] both argue such one-time costs belong in the ledger and must be amortized against recurring savings. No study in this corpus performs that amortization; "likely favorable" at high volume is a prediction, not a measurement, and its break-even point is unknown.

Gap 6: Reporting provenance is inconsistently observed. [25] argued for systematic reporting of the provenance of energy and carbon figures. The classification in Table 4 was constructible only by inference from each study's framing, which is itself evidence that this recommendation is not uniformly observed.

4.3 Discussion

For a practitioner selecting a quantization strategy where sustainability is a stated deployment priority, the reviewed literature supports several implications, each stated at the strength the evidence permits—in several cases weaker than the framing commonly attached to quantization.

The accuracy question is answerable from the literature; the environmental question largely is not. A practitioner asking whether a PTQ method preserves task capability at a given bit width can consult a mature evaluation literature [14, 15] and obtain a defensible answer; a practitioner asking how much energy the same choice saves will find protocols [17, 25, 28] but not transferable results. The appropriate response is to measure locally using those protocols rather than to cite a compression ratio as though it were an energy ratio.

Memory reduction and energy reduction are related but not equivalent. The relationship is mediated by hardware

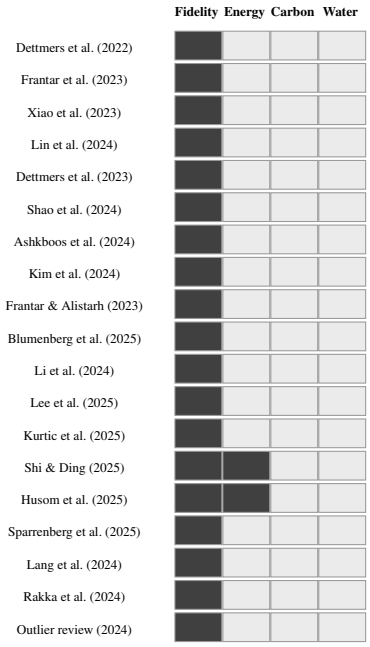


Figure 2. Coverage of the reviewed PTQ-focused corpus (Table 3) across the fidelity and sustainability axes. Dark cells indicate the axis was reported; light cells indicate NR. Fidelity is reported by nearly every study; energy is reported only by [17] and [3]; carbon and water are not reported in the reviewed PTQ-focused corpus. Rendered as a vector graphic (resolution-independent, exceeding 300 dpi at any print size).

support (Table 1), by whether the workload is bandwidth- or compute-bound [40], and by the interaction of power and time (Eq. 10). A format that halves storage does not thereby halve energy, and a fidelity-preserving strategy with dequantization overhead can consume more energy than a coarser one.

Task selection changes the answer. [15] report that quantization impact interacts with task difficulty and model size. Combined with the tokens-per-joule reasoning (Eq. 13), this implies that per-token efficiency gains can be offset if a quantized model requires longer or repeated generations to reach an acceptable answer.

Derived environmental figures should be reported as derived. Where a practitioner reports carbon or water savings from quantization, Section 3.2 and [25] indicate that the reported quantity is a scaled energy measurement, and the carbon-intensity or WUE factor used should accompany it.

One-time costs belong in the ledger. Where a method requires calibration or precision search [9, 19], the Green AI accounting principle [21, 26] requires that this cost be amortized against recurring savings.

Finally, and framed strictly as a proposition the literature has not tested: the reviewed corpus contains no study establishing a general accuracy–energy Pareto frontier across PTQ formats. Any such curve should be treated as a hypothesis to be tested rather than a relationship the literature has confirmed, and this review therefore presents no such curve (Figs. 1 and 2 remain illustrative rather than confirmatory).

4.4 Limitations of This Review

Search scope and corpus construction. The corpus was assembled from a predefined reference set rather than through an exhaustive database query with pre-registered inclusion criteria of the kind [33] describe. Relevant work outside the set was excluded by construction, and no formal assessment of coverage completeness was performed; reported frequencies and patterns should be interpreted as properties of the reviewed corpus rather than estimates of prevalence across the entire literature.

Depth of source records. The most significant limitation is that the bibliographic records available supplied titles, authors, and venues but not, for most entries, abstracts or result tables. Consequently, no numeric result is attributed to any reviewed paper, method descriptions in Section 2 are drawn from each work's own framing, and the NR cells in Tables 3–4 indicate non-recoverability from the record consulted rather than confirmed absence

in the primary source. Every table cell requires verification against the full text before this review is treated as a settled account of the field.

Bibliographic uncertainty. One candidate source could not be resolved through the accessible indexes and was therefore omitted rather than cited speculatively. One included entry [2] could not be assigned a verified author block from the records consulted.

Publication bias toward positive results. The PTQ literature is more likely to document methods and configurations that achieve acceptable quality or demonstrate measurable efficiency gains; configurations in which quantization degraded quality unacceptably, or expected efficiency gains did not materialize on real hardware, may be under-represented.

Rapid field evolution. Several entries are recent preprints [3, 13, 15–17, 30]. Hardware support for low-precision arithmetic—the mediating variable in nearly every efficiency claim reviewed here—changes across accelerator generations, so conclusions about realized efficiency are time-indexed.

Inability to verify reported or vendor-supplied figures. This review performed no independent measurement. Energy and power figures in the reviewed literature depend on instrumentation methodology and, in datacenter contexts, on operator-supplied efficiency factors that external researchers cannot audit [23, 25, 29]. Any carbon or water estimates derived from such measurements inherit the uncertainty of the underlying energy accounting (Section 3.2).

5. Conclusion and Future Scope

This review synthesized forty-three publications spanning post-training quantization algorithm design, quantization evaluation, and the sustainability accounting of machine learning, organized around two evaluation axes and the metrics each axis employs.

The central finding is structural rather than empirical. PTQ is routinely motivated as an efficiency and, by extension, sustainability intervention, and the reviewed algorithm papers [4–7, 9–11, 13] constitute a mature body of work on the fidelity question. The environmental question is addressed by a separate literature [20, 22, 23, 25, 27–30] that has developed its own metrics and protocols but that rarely takes a specific compression method as its unit of analysis. The intersection is occupied, in this corpus, by a small number of studies [1, 3, 17], concentrated at edge and commodity-hardware scale.

Four consequences follow. Energy evaluation of PTQ methods is sparse relative to accuracy evaluation and cannot be supplied retrospectively, since carbon and water are derived from energy measurements that were never taken. FP4 and NF4—formats of identical bit width but different level placement—have not been jointly compared on both axes. Server-scale sustainability comparison of quantization formats remains open, and edge findings do not extrapolate to it. Water consumption is absent from compression evaluation entirely.

Future scope. The six gaps identified in Section 4.2 constitute a direct research agenda. Priority should go to (i) a controlled, same-hardware energy benchmark of established PTQ methods (GPTQ, AWQ, SmoothQuant, OmniQuant, QuaRot) using the tokens-per-joule and latency instrumentation formalized in Section 3.2; (ii) a joint FP4-versus-NF4 comparison that reports fidelity (perplexity, benchmark accuracy) and energy on identical hardware, isolating the effect of level placement from bit width; (iii) extension of energy-aware quantization studies from edge to server-scale, batched-serving conditions, where bandwidth amortization and datacenter cooling introduce dynamics absent at the edge; (iv) first-of-their-kind water-footprint measurements for specific compression methods, following the WUE methodology of [29] and [30]; (v) explicit amortization accounting for calibration and precision-search overhead against recurring inference savings; and (vi) uniform provenance reporting—measured versus conversion-factor-derived—for every energy, carbon, and water figure published alongside a PTQ method. The question posed by this review's title admits no general answer from the reviewed literature: compression reliably reduces memory footprint, but whether, and by how much, it reduces energy, carbon, and water in a given deployment is a question the reviewed corpus supplies protocols for measuring but has not yet substantially answered.

References

[1] Sparrenberg, L., Deußer, T., Berger, A., & Sifa, R. (2025). Small and fast LLMs on commodity hardware: Post-training quantization in llama.cpp. In *Proceedings of the IEEE 12th International Conference on Data Science and Advanced Analytics (DSAA)*. https://doi.org/10.1109/DSAA65442.2025.11247985

[2] Addressing activation outliers in LLMs: A systematic review of post-training quantization techniques. *IEEE Xplore*, Document No. 10994764. (Author record not verifiable from the source consulted.)

[3] Husom, E. J., Goknil, A., Astekin, M., Shar, L. K., Kasen, A., Sen, S., Mithassel, B. Å., & Soylu, A. (2025). Sustainable LLM inference for edge AI: Evaluating quantized LLMs for energy efficiency, output accuracy, and inference latency. *arXiv preprint arXiv:2504.03360*.

[4] Dettmers, T., Lewis, M., Belkada, Y., & Zettlemoyer, L. (2022). LLM.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2208.07339.

[5] Frantar, E., Ashkboos, S., Hoefler, T., & Alistarh, D. (2023). GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations (ICLR)*. arXiv:2210.17323.

[6] Xiao, G., Lin, J., Seznec, M., Wu, H., Demouth, J., & Han, S. (2023). SmoothQuant: Accurate and efficient post-training quantization for large language models. In *International Conference on Machine Learning (ICML)*. arXiv:2211.10438.

[7] Lin, J., Tang, J., Tang, H., Yang, S., Dang, X., Gan, C., & Han, S. (2024). AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration. In *Proceedings of Machine Learning and Systems (MLSys)*. arXiv:2306.00978.

[8] Dettmers, T., Pagnoni, A., Holtzman, A., & Zettlemoyer, L. (2023). QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2305.14314.

[9] Shao, W., Chen, M., Zhang, Z., Xu, P., Zhao, L., Li, Z., Zhang, K., Gao, P., Qiao, Y., & Luo, P. (2024). OmniQuant: Omnidirectionally calibrated quantization for large language models. In *International Conference on Learning Representations (ICLR)*. arXiv:2308.13137.

[10] Ashkboos, S., Mohtashami, A., Croci, M. L., Li, B., Cameron, P., Jaggi, M., Alistarh, D., Hoefler, T., & Hensman, J. (2024). QuaRot: Outlier-free 4-bit inference in rotated LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2404.00456.

[11] Kim, S., Hooper, C., Gholami, A., Dong, Z., Li, X., Shen, S., Mahoney, M. W., & Keutzer, K. (2024). SqueezeLLM: Dense-and-sparse quantization. In *International Conference on Machine Learning (ICML)*.

[12] Frantar, E., & Alistarh, D. (2023). SparseGPT: Massive language models can be accurately pruned in one shot. In *International Conference on Machine Learning (ICML)*.

[13] Blumenberg, P., Graave, T., & Fingscheidt, T. (2025). Improving block-wise LLM quantization by 4-bit block-wise optimal float (BOF4): Analysis and variations. *arXiv preprint*.

[14] Li, S., Ning, X., Wang, L., Liu, T., Shi, X., Yan, S., Dai, G., Yang, H., & Wang, Y. (2024). Evaluating quantized large language models. *arXiv preprint arXiv:2402.18158*.

[15] Lee, J., Park, S., Kwon, J., Oh, J., & Kwon, Y. (2025). Exploring the trade-offs: Quantization methods, task difficulty, and model size in large language models from edge to giant. *arXiv preprint arXiv:2409.11055*.

[16] Kurtic, E., Marques, A., Pandit, S., Kurtz, M., & Alistarh, D. (2025). "Give me BF16 or give me death"? Accuracy–performance trade-offs in LLM quantization. *arXiv preprint arXiv:2411.02355*.

[17] Shi, T., & Ding, Y. (2025). Systematic characterization of LLM quantization: A performance, energy, and quality perspective. *arXiv preprint arXiv:2508.16712*.

[18] Lang, J., Guo, Z., & Huang, S. (2024). A comprehensive study on quantization techniques for large language models. In *Proceedings of the 4th International Conference on Artificial Intelligence, Robotics, and Communication (ICAIRC)*. IEEE.

[19] Rakka, M., Fouda, M. E., Khargonekar, P., & Kurdahi, F. (2024). A review of state-of-the-art mixed-precision neural network frameworks. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 46*(12).

[20] Strubell, E., Ganesh, A., & McCallum, A. (2019). Energy and policy considerations for deep learning in NLP. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)* (pp. 3645–3650).

[21] Schwartz, R., Dodge, J., Smith, N. A., & Etzioni, O. (2020). Green AI. *Communications of the ACM, 63*(12), 54–63.

[22] Lacoste, A., Luccioni, A., Schmidt, V., & Dandres, T. (2019). Quantifying the carbon emissions of machine learning. *arXiv preprint arXiv:1910.09700*.

[23] Patterson, D., Gonzalez, J., Le, Q., Liang, C., Munguia, L.-M., Rothchild, D., So, D., Texier, M., & Dean, J. (2021). Carbon emissions and large neural network training. *arXiv preprint arXiv:2104.10350*.

[24] Patterson, D., et al. (2022). The carbon footprint of machine learning training will plateau, then shrink. *Computer, 55*(7).

[25] Henderson, P., Hu, J., Romoff, J., Brunskill, E., Jurafsky, D., & Pineau, J. (2020). Towards the systematic reporting of the energy and carbon footprints of machine learning. *Journal of Machine Learning Research, 21*(248).

[26] Luccioni, A. S., Viguier, S., & Ligozat, A.-L. (2023). Estimating the carbon footprint of BLOOM, a 176B parameter language model. *Journal of Machine Learning Research, 24*(253).

[27] Luccioni, A. S., Jernite, Y., & Strubell, E. (2024). Power hungry processing: Watts driving the cost of AI deployment? In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)* (pp. 85–99).

[28] Samsi, S., Zhao, D., McDonald, J., Li, B., Michaleas, A., Jones, M., Bergeron, W., Kepner, J., Tiwari, D., & Gadepally, V. (2023). From words to watts: Benchmarking the energy costs of large language model inference. In *Proceedings of the IEEE High Performance Extreme Computing Conference (HPEC)* (pp. 1–9).

[29] Li, P., Yang, J., Islam, M. A., & Ren, S. (2023). Making AI less "thirsty": Uncovering and addressing the secret water footprint of AI models. *arXiv preprint arXiv:2304.03271*. Also in *Communications of the ACM, 68*(7), 2025.

[30] Jegham, N., Abdelatti, M., Koh, C. Y., Elmoubarki, L., & Hendawi, A. (2025). How hungry is AI? Benchmarking energy, water, and carbon footprint of LLM inference. *arXiv preprint arXiv:2505.09598*.

[31] Thompson, N. C., Greenewald, K., Lee, K., & Manso, G. F. (2021). Deep learning's diminishing returns: The cost of improvement is becoming unsustainable. *IEEE Spectrum, 58*(10), 50–55.

[32] Desislavov, R., Martínez-Plumed, F., & Hernández-Orallo, J. (2023). Trends in AI inference energy consumption: Beyond the performance-vs-parameter laws of deep learning. *Sustainable Computing: Informatics and Systems, 38*.

[33] Verdecchia, R., Sallou, J., & Cruz, L. (2023). A systematic review of Green AI. *WIREs Data Mining and Knowledge Discovery, 13*(4), e1507.

[34] Electric Power Research Institute. (2025). *Scaling intelligence: The exponential growth of AI's power needs* (EPRI White Paper).

[35] Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*.

[36] LeCun, Y., Denker, J., & Solla, S. (1989). Optimal brain damage. In *Advances in Neural Information Processing Systems (NeurIPS)*.

[37] He, Y., et al. (2024). A survey on deep neural network pruning: Taxonomy, comparison, analysis, and recommendations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

[38] Xu, X., Li, M., Tao, C., et al. (2024). A survey on knowledge distillation of large language models. *arXiv preprint arXiv:2402.13116*.

[39] Zhu, X., Li, J., Liu, Y., Ma, C., & Wang, W. (2024). A survey on model compression for large language models. *Transactions of the Association for Computational Linguistics, 12*, 1556–1577.

[40] Wan, Z., Wang, X., Liu, C., Alam, S., Zheng, Y., et al. (2024). Efficient large language models: A survey. *Transactions on Machine Learning Research*.

[41] Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., & Steinhardt, J. (2021). Measuring massive multitask language understanding. In *International Conference on Learning Representations (ICLR)*. arXiv:2009.03300.

[42] Cobbe, K., Kosaraju, V., Bavarian, M., et al. (2021). Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

[43] Merity, S., Xiong, C., Bradbury, J., & Socher, R. (2016). Pointer sentinel mixture models. *arXiv preprint arXiv:1609.07843*.